

Double Integrals in Polar Coordinates

Complete Worked Solutions – Exercise Set 14.3

Textbook: *Calculus*, 10th ed., Anton, Bivens & Davis

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Problems 1–10, 23–26, 27–34

Master formula used throughout:

$$\iint_R f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

Every problem below reduces to identifying the region R , writing its polar description ($\alpha \leq \theta \leq \beta$, $r_1(\theta) \leq r \leq r_2(\theta)$), and then integrating – inner integral in r first, then outer integral in θ . Never forget the extra factor of r that comes from $dA = r \, dr \, d\theta$.

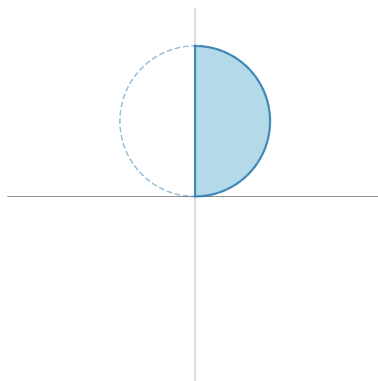
Part A – Problems 1–6: Evaluate the Iterated Integral

These are already set up in polar form. The strategy is always the same: integrate with respect to r first (treating θ as constant), simplify, then integrate with respect to θ .

Problem 1

$$\int_0^{\pi/2} \int_0^{\sin \theta} r \cos \theta \, dr \, d\theta$$

Region: $0 \leq \theta \leq \frac{\pi}{2}$, $0 \leq r \leq \sin \theta$. Since $r = \sin \theta$ is the circle $x^2 + y^2 = y$ (a circle of radius $\frac{1}{2}$ centered at $(0, \frac{1}{2})$), this is the upper-right quarter of that circle (the part in the first quadrant).



Step 1 – inner integral (in r): treat $\cos \theta$ as a constant.

$$\int_0^{\sin \theta} r \cos \theta \, dr = \cos \theta \left[\frac{r^2}{2} \right]_0^{\sin \theta} = \frac{\cos \theta \sin^2 \theta}{2}$$

Step 2 – outer integral (in θ):

$$\int_0^{\pi/2} \frac{\sin^2 \theta \cos \theta}{2} \, d\theta$$

Let $u = \sin \theta$, $du = \cos \theta \, d\theta$. When $\theta = 0$, $u = 0$; when $\theta = \pi/2$, $u = 1$.

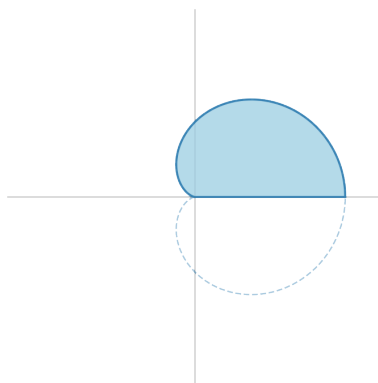
$$= \frac{1}{2} \int_0^1 u^2 \, du = \frac{1}{2} \left[\frac{u^3}{3} \right]_0^1 = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$

Answer: $\frac{1}{6}$

Problem 2

$$\int_0^{\pi} \int_0^{1+\cos \theta} r \, dr \, d\theta$$

Region: the upper half ($0 \leq \theta \leq \pi$) of the cardioid $r = 1 + \cos \theta$.



Step 1 – inner integral:

$$\int_0^{1+\cos\theta} r \, dr = \left[\frac{r^2}{2} \right]_0^{1+\cos\theta} = \frac{(1+\cos\theta)^2}{2}$$

Step 2 – expand and integrate: $(1 + \cos\theta)^2 = 1 + 2\cos\theta + \cos^2\theta$, and recall $\int_0^\pi \cos^2\theta \, d\theta = \frac{\pi}{2}$ (half-angle identity).

$$\frac{1}{2} \int_0^\pi (1 + 2\cos\theta + \cos^2\theta) \, d\theta = \frac{1}{2} \left[\theta + 2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^\pi$$

Evaluating: $\theta = \pi \Rightarrow \pi + 0 + \frac{\pi}{2} + 0 = \frac{3\pi}{2}$, and at $\theta = 0$ everything is 0.

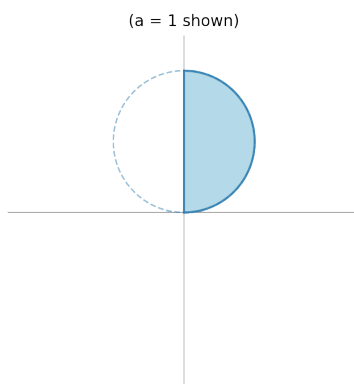
$$= \frac{1}{2} \cdot \frac{3\pi}{2} = \frac{3\pi}{4}$$

Answer: $\frac{3\pi}{4}$

Problem 3

$$\int_0^{\pi/2} \int_0^{a \sin\theta} r^2 \, dr \, d\theta$$

Region: first-quadrant half of the circle $r = a \sin\theta$ (diameter a , tangent to the x -axis at the origin).



Step 1 – inner integral:

$$\int_0^{a \sin\theta} r^2 \, dr = \left[\frac{r^3}{3} \right]_0^{a \sin\theta} = \frac{a^3 \sin^3\theta}{3}$$

Step 2 – outer integral: using the reduction formula $\int_0^{\pi/2} \sin^3 \theta \, d\theta = \frac{2}{3}$,

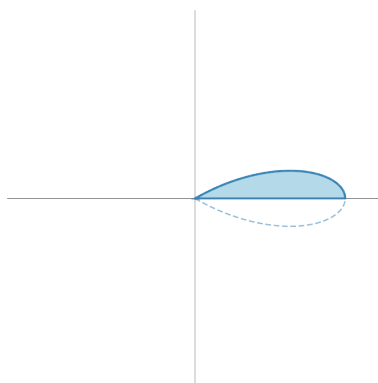
$$\frac{a^3}{3} \int_0^{\pi/2} \sin^3 \theta \, d\theta = \frac{a^3}{3} \cdot \frac{2}{3} = \frac{2a^3}{9}$$

Answer: $\frac{2a^3}{9}$

Problem 4

$$\int_0^{\pi/6} \int_0^{\cos 3\theta} r \, dr \, d\theta$$

Region: one half of a single petal of the rose $r = \cos 3\theta$ (the full petal runs from $\theta = -\pi/6$ to $\pi/6$; here we take just $0 \leq \theta \leq \pi/6$).



Step 1 – inner integral:

$$\int_0^{\cos 3\theta} r \, dr = \frac{\cos^2 3\theta}{2}$$

Step 2 – outer integral: substitute $u = 3\theta$, $du = 3 \, d\theta$; limits $0 \rightarrow \pi/2$.

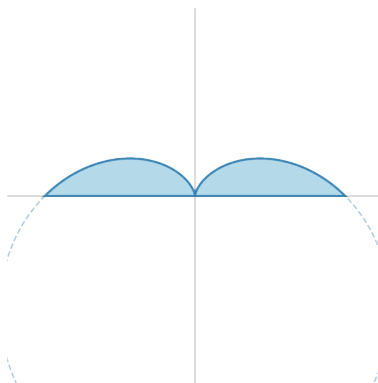
$$\frac{1}{2} \int_0^{\pi/6} \cos^2 3\theta \, d\theta = \frac{1}{2} \cdot \frac{1}{3} \int_0^{\pi/2} \cos^2 u \, du = \frac{1}{6} \cdot \frac{\pi}{4} = \frac{\pi}{24}$$

Answer: $\frac{\pi}{24}$

Problem 5

$$\int_0^{\pi} \int_0^{1-\sin \theta} r^2 \cos \theta \, dr \, d\theta$$

Region: upper half of the cardioid $r = 1 - \sin \theta$ (dimpled downward, symmetric about the y -axis).



Step 1 – inner integral:

$$\int_0^{1-\sin\theta} r^2 dr = \frac{(1-\sin\theta)^3}{3}$$

Step 2 – outer integral:

$$\frac{1}{3} \int_0^\pi (1-\sin\theta)^3 \cos\theta d\theta$$

Let $u = 1 - \sin\theta$, $du = -\cos\theta d\theta$. An antiderivative in θ is $-\frac{(1-\sin\theta)^4}{4}$. Evaluate at the endpoints:

$$\theta = \pi : -\frac{(1-0)^4}{4} = -\frac{1}{4} \qquad \theta = 0 : -\frac{(1-0)^4}{4} = -\frac{1}{4}$$

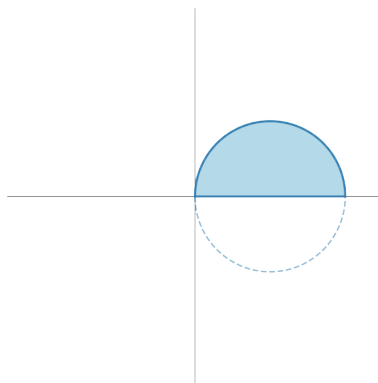
Both endpoints give the same value (since $\sin 0 = \sin \pi = 0$), so the difference is 0.

$$\frac{1}{3} \left(-\frac{1}{4} - \left(-\frac{1}{4} \right) \right) = \frac{1}{3}(0) = 0$$

Answer: 0
Problem 6

$$\int_0^{\pi/2} \int_0^{\cos\theta} r^3 dr d\theta$$

Region: the right half ($0 \leq \theta \leq \pi/2$) of the circle $r = \cos\theta$.

**Step 1 – inner integral:**

$$\int_0^{\cos\theta} r^3 dr = \frac{\cos^4\theta}{4}$$

Step 2 – outer integral: using $\int_0^{\pi/2} \cos^4\theta d\theta = \frac{3\pi}{16}$,

$$\frac{1}{4} \cdot \frac{3\pi}{16} = \frac{3\pi}{64}$$

Answer: $\frac{3\pi}{64}$

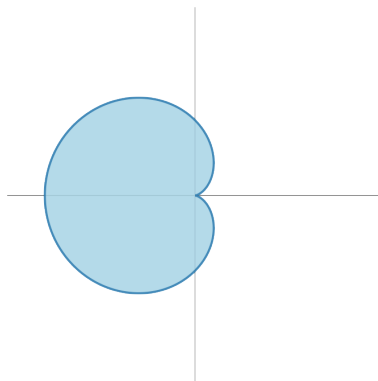
Part B – Problems 7–10: Areas via Polar Double Integrals

The area formula is a special case of the master formula with $f = 1$:

$$\text{Area} = \iint_R dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} r \, dr \, d\theta$$

Problem 7 – Area enclosed by the cardioid $r = 1 - \cos \theta$

Region: the full cardioid, so $0 \leq \theta \leq 2\pi$, $0 \leq r \leq 1 - \cos \theta$.



$$A = \int_0^{2\pi} \int_0^{1-\cos \theta} r \, dr \, d\theta = \int_0^{2\pi} \frac{(1 - \cos \theta)^2}{2} d\theta$$

Expand: $(1 - \cos \theta)^2 = 1 - 2 \cos \theta + \cos^2 \theta$. Over a full period, $\int_0^{2\pi} \cos \theta \, d\theta = 0$ and $\int_0^{2\pi} \cos^2 \theta \, d\theta = \pi$, so

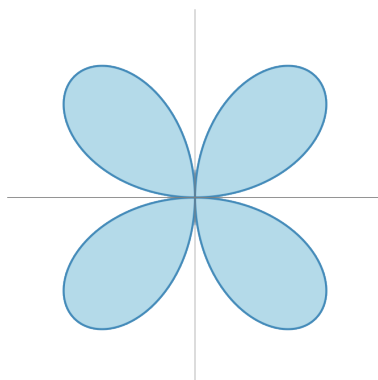
$$\int_0^{2\pi} (1 - 2 \cos \theta + \cos^2 \theta) d\theta = 2\pi - 0 + \pi = 3\pi$$

$$A = \frac{1}{2}(3\pi) = \frac{3\pi}{2}$$

Answer: $\frac{3\pi}{2}$

Problem 8 – Area enclosed by the rose $r = \sin 2\theta$

Region: $r = \sin 2\theta$ is a *four-petaled* rose (since $n = 2$ is even, the rose has $2n = 4$ petals). One petal lies in the first quadrant, traced as θ runs from 0 to $\pi/2$ (where $\sin 2\theta \geq 0$).



One petal:

$$A_{\text{petal}} = \int_0^{\pi/2} \int_0^{\sin 2\theta} r \, dr \, d\theta = \int_0^{\pi/2} \frac{\sin^2 2\theta}{2} \, d\theta$$

Using $\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$:

$$= \frac{1}{4} \int_0^{\pi/2} (1 - \cos 4\theta) \, d\theta = \frac{1}{4} \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{\pi/2} = \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{8}$$

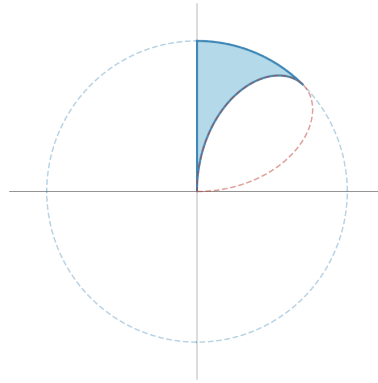
All four petals are congruent, so the total area is

$$A = 4 \times \frac{\pi}{8} = \frac{\pi}{2}$$

Answer: $\frac{\pi}{2}$

Problem 9 – Region in Q1 bounded by $r = 1$ and $r = \sin 2\theta$, $\pi/4 \leq \theta \leq \pi/2$

Region: On $[\frac{\pi}{4}, \frac{\pi}{2}]$ the rose petal $r = \sin 2\theta$ shrinks from 1 (at $\theta = \pi/4$, where it touches the circle) down to 0 (at $\theta = \pi/2$), staying entirely *inside* the circle $r = 1$. So the region is the thin sliver between the circle (outer) and the petal (inner): $\sin 2\theta \leq r \leq 1$.



$$A = \int_{\pi/4}^{\pi/2} \int_{\sin 2\theta}^1 r \, dr \, d\theta = \int_{\pi/4}^{\pi/2} \frac{1 - \sin^2 2\theta}{2} \, d\theta$$

Using $\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$, we get $1 - \sin^2 2\theta = \frac{1 + \cos 4\theta}{2}$:

$$= \frac{1}{4} \int_{\pi/4}^{\pi/2} (1 + \cos 4\theta) \, d\theta = \frac{1}{4} \left[\theta + \frac{\sin 4\theta}{4} \right]_{\pi/4}^{\pi/2}$$

At $\theta = \pi/2$: $\frac{\pi}{2} + \frac{\sin 2\pi}{4} = \frac{\pi}{2}$. At $\theta = \pi/4$: $\frac{\pi}{4} + \frac{\sin \pi}{4} = \frac{\pi}{4}$.

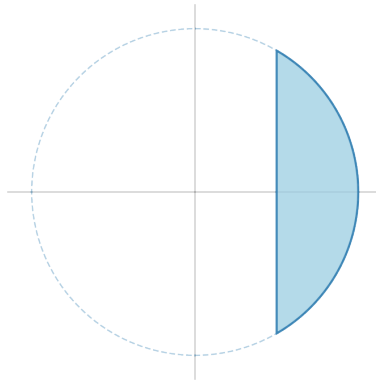
$$A = \frac{1}{4} \left(\frac{\pi}{2} - \frac{\pi}{4} \right) = \frac{1}{4} \cdot \frac{\pi}{4} = \frac{\pi}{16}$$

Answer: $\frac{\pi}{16}$

Problem 10 – Area inside the circle $x^2 + y^2 = 4$ and to the right of $x = 1$

Convert boundaries to polar: the circle is $r = 2$. The line $x = 1$ becomes $r \cos \theta = 1$, i.e. $r = \sec \theta$.

Intersection: $\sec \theta = 2 \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \pm \frac{\pi}{3}$. For $-\pi/3 \leq \theta \leq \pi/3$, the region runs from the line out to the circle: $\sec \theta \leq r \leq 2$.



By symmetry about the x -axis, integrate over $[0, \pi/3]$ and double:

$$A = 2 \int_0^{\pi/3} \int_{\sec \theta}^2 r \, dr \, d\theta = 2 \int_0^{\pi/3} \frac{4 - \sec^2 \theta}{2} \, d\theta = \int_0^{\pi/3} (4 - \sec^2 \theta) \, d\theta$$

Recall $\int \sec^2 \theta \, d\theta = \tan \theta$:

$$= \left[4\theta - \tan \theta \right]_0^{\pi/3} = \left(\frac{4\pi}{3} - \tan \frac{\pi}{3} \right) - 0 = \frac{4\pi}{3} - \sqrt{3}$$

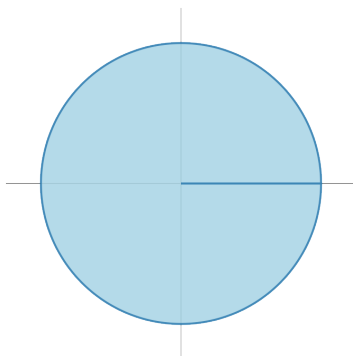
Answer: $\frac{4\pi}{3} - \sqrt{3}$

Part C – Problems 23–26: Evaluate $\iint_R f(x, y) dA$ Using Polar Coordinates

Here the region and integrand are given in (x, y) -form; we translate both into polar using $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$, $dA = r dr d\theta$.

Problem 23 – $\iint_R \sin(x^2 + y^2) dA$, R : disk $x^2 + y^2 \leq 9$

Region: full disk of radius 3: $0 \leq \theta \leq 2\pi$, $0 \leq r \leq 3$.



$$\iint_R \sin(x^2 + y^2) dA = \int_0^{2\pi} \int_0^3 \sin(r^2) r dr d\theta$$

Inner integral: let $u = r^2$, $du = 2r dr$:

$$\int_0^3 \sin(r^2) r dr = \frac{1}{2} \int_0^9 \sin u du = \frac{1}{2} [-\cos u]_0^9 = \frac{1 - \cos 9}{2}$$

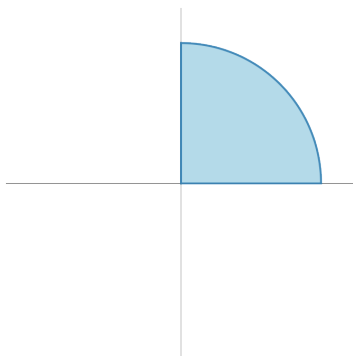
Outer integral:

$$\int_0^{2\pi} \frac{1 - \cos 9}{2} d\theta = 2\pi \cdot \frac{1 - \cos 9}{2} = \pi(1 - \cos 9)$$

Answer: $\pi(1 - \cos 9)$

Problem 24 – $\iint_R \sqrt{9 - x^2 - y^2} dA$, R : Q1 part of disk $x^2 + y^2 \leq 9$

Region: quarter disk of radius 3: $0 \leq \theta \leq \pi/2$, $0 \leq r \leq 3$.



$$\iint_R \sqrt{9 - x^2 - y^2} dA = \int_0^{\pi/2} \int_0^3 \sqrt{9 - r^2} r dr d\theta$$

Inner integral: let $u = 9 - r^2$, $du = -2r dr$:

$$\int_0^3 r\sqrt{9-r^2} dr = \frac{1}{2} \int_0^9 \sqrt{u} du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} \Big|_0^9 = \frac{1}{3} (27) = 9$$

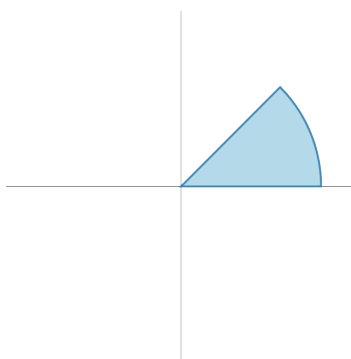
Outer integral:

$$\int_0^{\pi/2} 9 d\theta = 9 \cdot \frac{\pi}{2} = \frac{9\pi}{2}$$

$$\text{Answer: } \frac{9\pi}{2}$$

Problem 25 – $\iint_R \frac{1}{1+x^2+y^2} dA$, R : Q1 sector bounded by $y=0$, $y=x$, $x^2+y^2=4$

Region: $y=0$ is $\theta=0$; $y=x$ is $\theta=\pi/4$; the circle is $r=2$. So $0 \leq \theta \leq \pi/4$, $0 \leq r \leq 2$.



$$\iint_R \frac{1}{1+x^2+y^2} dA = \int_0^{\pi/4} \int_0^2 \frac{r}{1+r^2} dr d\theta$$

Inner integral:

$$\int_0^2 \frac{r}{1+r^2} dr = \frac{1}{2} \left[\ln(1+r^2) \right]_0^2 = \frac{1}{2} \ln 5$$

Outer integral:

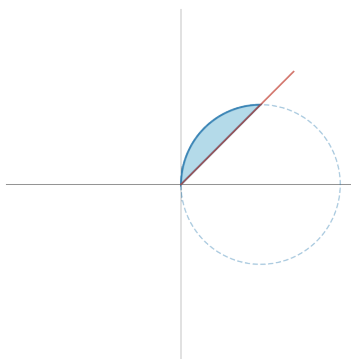
$$\int_0^{\pi/4} \frac{\ln 5}{2} d\theta = \frac{\pi}{4} \cdot \frac{\ln 5}{2} = \frac{\pi \ln 5}{8}$$

$$\text{Answer: } \frac{\pi \ln 5}{8}$$

Problem 26 – $\iint_R 2y dA$, R : Q1 region under circle $(x-1)^2 + y^2 = 1$ and above the line $y=x$

Convert the circle: $(x-1)^2 + y^2 = 1 \Rightarrow x^2 + y^2 = 2x \Rightarrow r^2 = 2r \cos \theta \Rightarrow r = 2 \cos \theta$ (a circle of radius 1 through the origin).

The line $y=x$ is $\theta=\pi/4$. Since we want the part of the disk that also lies *above* the line, θ runs from $\pi/4$ up to $\pi/2$ (where $r \rightarrow 0$), and for each such θ , r runs from 0 out to the circle: $0 \leq r \leq 2 \cos \theta$.



$$\iint_R 2y \, dA = \int_{\pi/4}^{\pi/2} \int_0^{2 \cos \theta} 2(r \sin \theta) r \, dr \, d\theta = \int_{\pi/4}^{\pi/2} 2 \sin \theta \int_0^{2 \cos \theta} r^2 \, dr \, d\theta$$

Inner integral:

$$\int_0^{2 \cos \theta} r^2 \, dr = \frac{(2 \cos \theta)^3}{3} = \frac{8 \cos^3 \theta}{3}$$

So the outer integral becomes

$$\int_{\pi/4}^{\pi/2} 2 \sin \theta \cdot \frac{8 \cos^3 \theta}{3} \, d\theta = \frac{16}{3} \int_{\pi/4}^{\pi/2} \sin \theta \cos^3 \theta \, d\theta$$

Let $u = \cos \theta$, $du = -\sin \theta \, d\theta$; an antiderivative is $-\frac{\cos^4 \theta}{4}$:

$$\frac{16}{3} \left[-\frac{\cos^4 \theta}{4} \right]_{\pi/4}^{\pi/2} = \frac{16}{3} \left(0 - \left(-\frac{(\frac{\sqrt{2}}{2})^4}{4} \right) \right) = \frac{16}{3} \cdot \frac{1}{16} = \frac{1}{3}$$

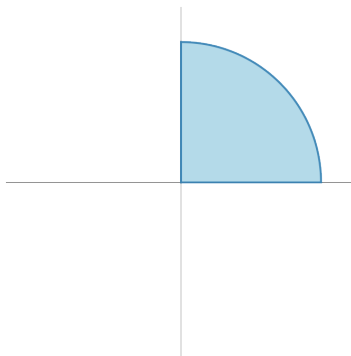
Answer: $\frac{1}{3}$

Part D – Problems 27–34: Convert Rectangular Iterated Integrals to Polar

The general strategy: (1) sketch the rectangular region from the limits, (2) rewrite its boundary curves in polar form, (3) rewrite the integrand using $x^2 + y^2 = r^2$, and (4) don't forget the extra r from $dA = r dr d\theta$.

Problem 27 – $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$

Region: $0 \leq x \leq 1$, $0 \leq y \leq \sqrt{1-x^2}$ is the quarter disk of radius 1 in Q1. So $0 \leq \theta \leq \pi/2$, $0 \leq r \leq 1$, and $x^2 + y^2 = r^2$.

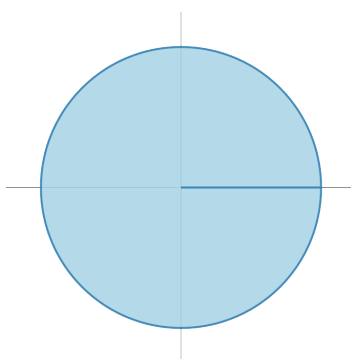


$$\int_0^{\pi/2} \int_0^1 r^2 \cdot r dr d\theta = \int_0^{\pi/2} \int_0^1 r^3 dr d\theta = \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{8}$$

Answer: $\frac{\pi}{8}$

Problem 28 – $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)} dx dy$

Region: full disk of radius 2 (for each $y \in [-2, 2]$, x spans the whole chord). So $0 \leq \theta \leq 2\pi$, $0 \leq r \leq 2$.



$$\int_0^{2\pi} \int_0^2 e^{-r^2} r dr d\theta$$

Inner integral: let $u = -r^2$, $du = -2r dr$:

$$\int_0^2 e^{-r^2} r dr = -\frac{1}{2} e^{-r^2} \Big|_0^2 = \frac{1 - e^{-4}}{2}$$

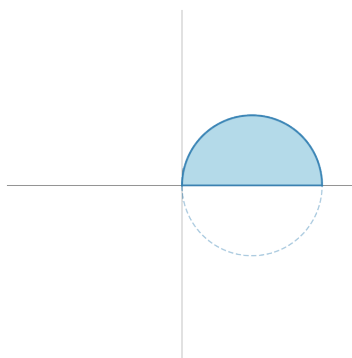
Outer integral:

$$\int_0^{2\pi} \frac{1 - e^{-4}}{2} d\theta = 2\pi \cdot \frac{1 - e^{-4}}{2} = \pi (1 - e^{-4})$$

Answer: $\pi(1 - e^{-4})$

Problem 29 – $\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2 + y^2} \, dy \, dx$

Region: $y = \sqrt{2x - x^2}$ means $y^2 = 2x - x^2$, i.e. $x^2 + y^2 = 2x$ – a circle of radius 1 centered at $(1, 0)$. With $y \geq 0$ and $x \in [0, 2]$, this is the *upper half* of that circle. In polar: $r^2 = 2r \cos \theta \Rightarrow r = 2 \cos \theta$, traced for $0 \leq \theta \leq \pi/2$.



$$\int_0^{\pi/2} \int_0^{2 \cos \theta} r \cdot r \, dr \, d\theta = \int_0^{\pi/2} \frac{(2 \cos \theta)^3}{3} \, d\theta = \frac{8}{3} \int_0^{\pi/2} \cos^3 \theta \, d\theta$$

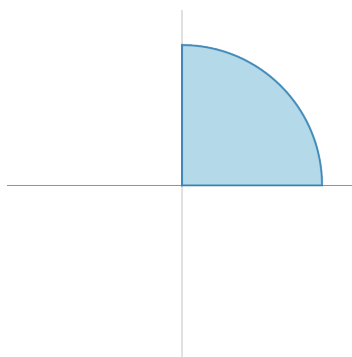
Using $\int_0^{\pi/2} \cos^3 \theta \, d\theta = \frac{2}{3}$:

$$= \frac{8}{3} \cdot \frac{2}{3} = \frac{16}{9}$$

Answer: $\frac{16}{9}$

Problem 30 – $\int_0^1 \int_0^{\sqrt{1-y^2}} \cos(x^2 + y^2) \, dx \, dy$

Region: quarter disk of radius 1 in Q1 (same shape as #27, just integrated in the other order). So $0 \leq \theta \leq \pi/2$, $0 \leq r \leq 1$.



$$\int_0^{\pi/2} \int_0^1 \cos(r^2) r \, dr \, d\theta$$

Inner integral: let $u = r^2$, $du = 2r \, dr$:

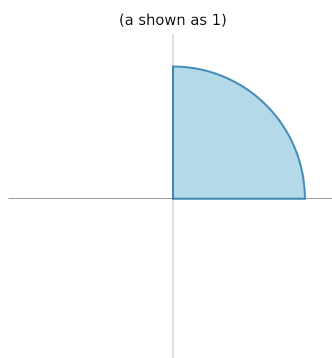
$$\int_0^1 \cos(r^2) r \, dr = \frac{1}{2} \int_0^1 \cos u \, du = \frac{1}{2} \sin(1)$$

Outer integral:

$$\int_0^{\pi/2} \frac{\sin 1}{2} d\theta = \frac{\pi}{2} \cdot \frac{\sin 1}{2} = \frac{\pi}{4} \sin 1$$

$$\text{Answer: } \frac{\pi}{4} \sin 1$$

$$\text{Problem 31} - \int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{1}{(1+x^2+y^2)^{3/2}} dy dx, \quad a > 0$$

Region: quarter disk of radius a in Q1: $0 \leq \theta \leq \pi/2$, $0 \leq r \leq a$.

$$\int_0^{\pi/2} \int_0^a \frac{r}{(1+r^2)^{3/2}} dr d\theta$$

Inner integral: let $u = 1 + r^2$, $du = 2r dr$; when $r = 0$, $u = 1$, when $r = a$, $u = 1 + a^2$:

$$\int_0^a \frac{r dr}{(1+r^2)^{3/2}} = \frac{1}{2} \int_1^{1+a^2} u^{-3/2} du = \left[-u^{-1/2} \right]_1^{1+a^2} = 1 - \frac{1}{\sqrt{1+a^2}}$$

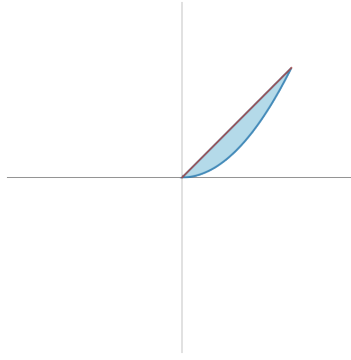
Outer integral:

$$\int_0^{\pi/2} \left(1 - \frac{1}{\sqrt{1+a^2}} \right) d\theta = \frac{\pi}{2} \left(1 - \frac{1}{\sqrt{1+a^2}} \right)$$

$$\text{Answer: } \frac{\pi}{2} \left(1 - \frac{1}{\sqrt{1+a^2}} \right)$$

$$\text{Problem 32} - \int_0^1 \int_y^{\sqrt{y}} \sqrt{x^2+y^2} dx dy$$

Region: $y \leq x \leq \sqrt{y}$ is equivalent to $x^2 \leq y \leq x$ – the region between the line $y = x$ (above) and the parabola $y = x^2$ (below), for $x \in [0, 1]$.**Line $y = x$:** $\theta = \pi/4$. **Parabola $y = x^2$:** $r \sin \theta = r^2 \cos^2 \theta \Rightarrow r = \frac{\sin \theta}{\cos^2 \theta} = \tan \theta \sec \theta$.For a fixed $\theta \in [0, \pi/4]$, a ray from the origin stays inside the region until it exits through the parabola, so $0 \leq r \leq \tan \theta \sec \theta$, with $0 \leq \theta \leq \pi/4$.



$$\int_0^{\pi/4} \int_0^{\tan \theta \sec \theta} r \cdot r \, dr \, d\theta = \frac{1}{3} \int_0^{\pi/4} \tan^3 \theta \sec^3 \theta \, d\theta$$

Evaluate the trig integral: write $\tan^3 \theta \sec^3 \theta = (\sec^2 \theta - 1) \sec^2 \theta \cdot (\sec \theta \tan \theta)$. Let $u = \sec \theta$, $du = \sec \theta \tan \theta \, d\theta$:

$$\int (u^2 - 1)u^2 \, du = \frac{u^5}{5} - \frac{u^3}{3} = \frac{\sec^5 \theta}{5} - \frac{\sec^3 \theta}{3}$$

Evaluate from 0 to $\pi/4$ (where $\sec 0 = 1$, $\sec \frac{\pi}{4} = \sqrt{2}$):

$$\left[\frac{\sec^5 \theta}{5} - \frac{\sec^3 \theta}{3} \right]_0^{\pi/4} = \left(\frac{4\sqrt{2}}{5} - \frac{2\sqrt{2}}{3} \right) - \left(\frac{1}{5} - \frac{1}{3} \right) = \frac{2\sqrt{2}}{15} + \frac{2}{15} = \frac{2(\sqrt{2} + 1)}{15}$$

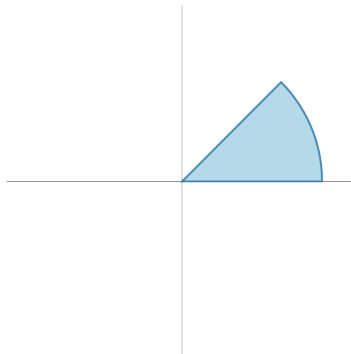
So the full integral is

$$\frac{1}{3} \cdot \frac{2(\sqrt{2} + 1)}{15} = \frac{2(\sqrt{2} + 1)}{45}$$

Answer: $\frac{2(\sqrt{2} + 1)}{45}$

Problem 33 – $\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} \frac{1}{\sqrt{1+x^2+y^2}} \, dx \, dy$

Region: x runs from the line $x = y$ out to the circle $x = \sqrt{4 - y^2}$ (i.e. $x^2 + y^2 = 4$), for $y \in [0, \sqrt{2}]$. At $y = 0$: $x \in [0, 2]$ (the whole radius along the x -axis, $\theta = 0$). At $y = \sqrt{2}$: $x = \sqrt{2} = y$ (a single point, where line meets circle at $\theta = \pi/4$). So the region is the pie-slice sector $0 \leq \theta \leq \pi/4$, $0 \leq r \leq 2$.



$$\int_0^{\pi/4} \int_0^2 \frac{r}{\sqrt{1+r^2}} \, dr \, d\theta$$

Inner integral: let $u = 1 + r^2$, $du = 2r \, dr$; $r = 0 \Rightarrow u = 1$, $r = 2 \Rightarrow u = 5$:

$$\int_0^2 \frac{r \, dr}{\sqrt{1+r^2}} = \frac{1}{2} \int_1^5 u^{-1/2} \, du = \left[u^{1/2} \right]_1^5 = \sqrt{5} - 1$$

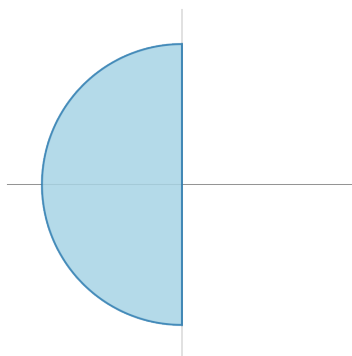
Outer integral:

$$\int_0^{\pi/4} (\sqrt{5} - 1) d\theta = \frac{\pi}{4}(\sqrt{5} - 1)$$

$$\text{Answer: } \frac{\pi}{4}(\sqrt{5} - 1)$$

$$\text{Problem 34} - \int_{-4}^0 \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} 3x \, dy \, dx$$

Region: $x \in [-4, 0]$ with y spanning the full chord $-\sqrt{16-x^2} \leq y \leq \sqrt{16-x^2}$ – this is the *left half* ($x \leq 0$) of the disk of radius 4. In polar, that is $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$, $0 \leq r \leq 4$.



$$\int_{\pi/2}^{3\pi/2} \int_0^4 3(r \cos \theta) r \, dr \, d\theta = 3 \left(\int_{\pi/2}^{3\pi/2} \cos \theta \, d\theta \right) \left(\int_0^4 r^2 \, dr \right)$$

Radial part:

$$\int_0^4 r^2 \, dr = \frac{64}{3}$$

Angular part:

$$\int_{\pi/2}^{3\pi/2} \cos \theta \, d\theta = \left[\sin \theta \right]_{\pi/2}^{3\pi/2} = \sin \frac{3\pi}{2} - \sin \frac{\pi}{2} = -1 - 1 = -2$$

Combine:

$$3 \times (-2) \times \frac{64}{3} = -128$$

$$\text{Answer: } -128$$